

Review

Sustainable farming practices and soil health: a pathway to achieving SDGs and future prospects

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Abstract

Sustainable farming practices aim to produce agricultural products at a low environmental cost, ensuring food availability for future generations. These approaches combine the production of agricultural crops and livestock for site-specific uses, with an emphasis on long-term objectives including satisfying the food supply worldwide, improving the environment, making the most efficient use of resources and boosting the welfare of farmers and society as a whole. Sustainability in farming is mostly dependent on soil health, which is the soil's capacity to function within ecological and land-use boundaries. It enhances general wellness, maintains the purity of the air and water, and supports biological productivity. The depletion of soil functions brought about by modern agricultural intensification has an impact on ecosystem services and long-term productivity. The goal of sustainable farming is to raise fertility and organic matter in the soil. Key practices include carbon farming, conservation agriculture, integrated pest management (IPM), and precision farming. In order to support sustainable crop production, these techniques improve soil fertility, structure, and biodiversity. Going forward, research should concentrate on developing novel approaches to tackle the issues of climate change and global food security.

Keywords Sustainable farming · Sustainable development goals (SDG's) · Soil health · Precision agriculture · Carbon farming · IPM · Global food security · Conservation farming · Soil biodiversity

1 Introduction

Sustainable farming refers to agricultural practices that enhances the use of non-renewable resources, helps to preserve and expand natural resources and protects the environment. Sustainable farming methods and practices offer ways to produce food and other agricultural products at a low environmental cost that have no detrimental effect on the availability and accessibility of food or the general well-being of future generations [1]. An integrated system of plant and animal production practices with a site-specific application is referred to as sustainable farming. Its long-term goals include meeting the world's food and fiber needs, improving the environment and the natural resource base that supports the agricultural economy (Fig. 1), maximizing the use of both renewable and non-renewable resources, integrating natural biological cycles and controls where appropriate, maintaining the viability of farm operations financially and improving the lives of farmers and society overall [2, 3].

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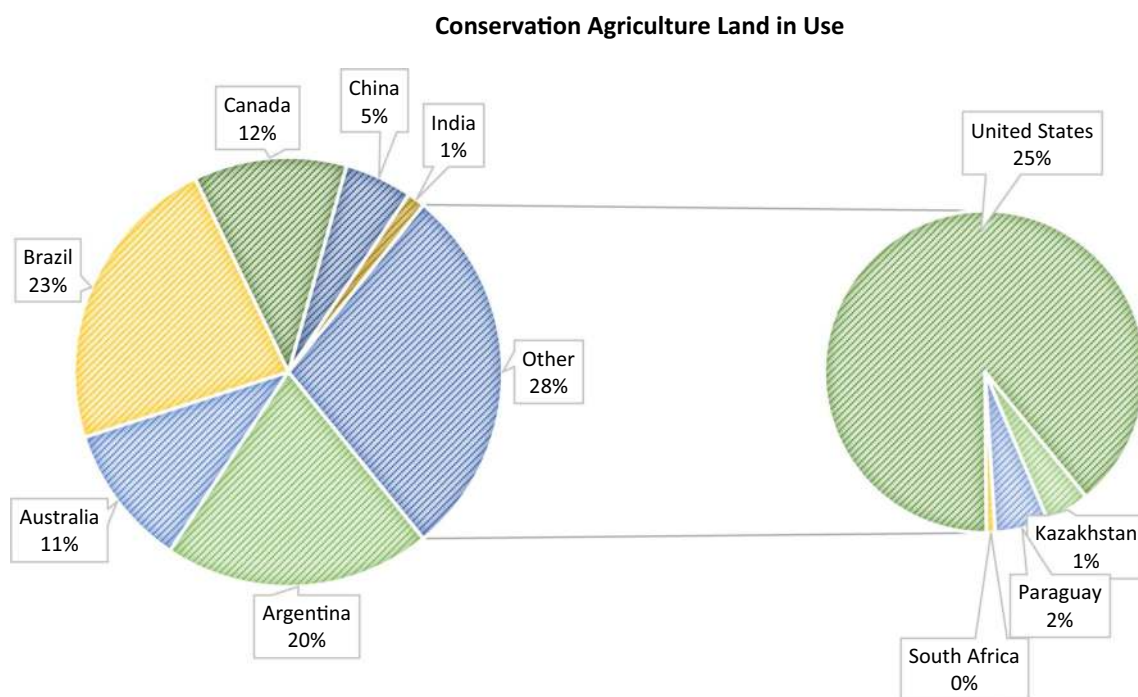


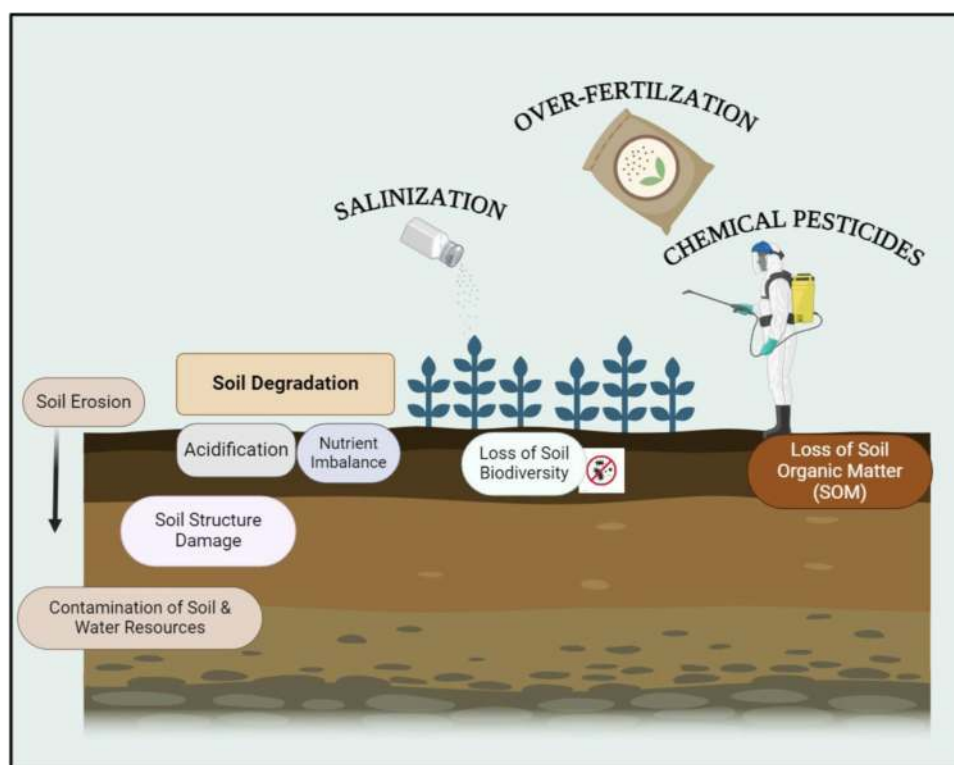
Fig. 1 The pie chart above shows the percentage of land used for conservation agriculture worldwide

According to definitions, soil health is the capacity of the soil to carry on as an essential living system within ecological and land-use boundaries, supporting biological productivity, preserving the quality of the air & water and enhancing the health of people, animals and plants [3, 4]. “The ability of the soil to sustain productivity, diversity and environmental services of terrestrial ecosystems” is the definition of soil health given by the Intergovernmental Technical Panel on Soils (ITPS) in a more current definition [4, 5]. The favorable physical (texture, water-holding capacity), chemical (pH, soil organic matter; SOM) and biological (microbial diversity, N mineralization and soil respiration) characteristics of healthy, productive crops are what are considered to be the hallmarks of healthy soil. It is believed that soil is a living, dynamic ecosystem with a diverse range of micro- and macro-biota that control its characteristics [4–6]. Modern technology-induced agricultural intensification has reduced soil’s ability to sustain its functions, which has an impact on long-term production and results in a loss of ecosystem services (Fig. 2). The main goal of sustainable farming techniques is to improve soil health by raising the amount of organic matter in the soil and enhancing its fertility and productivity [7].

Future agricultural land shortages and high food demands will necessitate increasing crop yields through sustainable methods. By incorporating their understanding of soil function into useful approaches that improve producers’ ability to assess the sustainability of their management practices, scientists can significantly contribute to the global sustainability of agricultural lands [8]. A major problem is meeting the expected demand for food production that is both sustainable and beneficial for health. In actuality, one of the main objectives of sustainable agriculture is raising crop output while preventing climate change and maintaining agroecosystems [5, 9]. However, in order to meet agricultural demand, an excessive amount of synthetic fertilizer and pesticides have been used, which has resulted in land degradation and environmental contamination in a number of agroecosystems, negatively affecting aquatic ecosystems, marine life, and people [1, 10–12]. Food security is linked to sustainable agriculture. To ensure food security, there must be sufficient food production or availability, access to food and the ability to purchase it, as well as safety, nutritional sufficiency (including energy, proteins, and micronutrients), and economic stability [8–13]. To solve the issue of long-term food security, innovative sustainable agricultural techniques and methods should be developed and implemented at all agricultural production scales. This article’s goal is to analyze and highlight different ideas and practices that are unique for all those involved, under the general heading of sustainable agriculture [14–17].

The restoration or improvement of soil health requires global attention. It is anticipated that the evaluation of soil health indicators will improve our comprehension of the underlying mechanisms supporting sustainable farming methods. In this review, research results on soil health management, sustainable farming methods, and their contribution to

Fig. 2 The graphic illustrates the different detrimental effects of chemical-based farming on various soil layers, demonstrating how these methods can result in contamination, nutrient imbalances, erosion, acidity of the soil, reduction of soil biodiversity, and degradation of the soil. Understanding the complexity and interconnectedness of the problems brought on by chemical-based farming methods is made easier with the aid of this visual representation



sustainable crop production will be covered. Its goal is to increase knowledge about how improved farming methods can improve soil health and how they relate to the Sustainable Development Goals.

2 Key sustainable farming practices

One potential solution to lessening the negative effects of agricultural activities on the environment is the use of sustainable agricultural technology. A growing number of people in the agricultural industry are realizing how urgent it is to implement more sustainable practices in order to evaluate their benefits to the environment [18]. According to MacRae et al. [19] achieving sustainability in agriculture depends on pursuing particular agricultural techniques meant to reduce the long-term effects of human activity on natural resources [19]. Of all the solutions available, integrated pest management (IPM) [20], precision farming [21], carbon farming [22] and conservation agriculture [23] are clearly the most successful. These provide creative answers to the problems associated with sustainable agriculture, protecting the environment and promising the production of healthy, high-quality food. Farmers that adopt these methods can address the pressing need for sustainable agricultural systems, encourage ecological resilience and protect natural resources [15, 23].

3 Carbon farming

Less-intensive and carefully planned farming techniques can help stabilize the agricultural output's reliance on climate change. To find farming methods that can manage the delicate balance between climatic change and agricultural productions, the agro-environmental characteristics must be carefully taken into account. In this scenario, carbon farming provides a comprehensive and sustainable approach to land-use management that benefits society and the environment [22, 23]. In recent years, carbon farming has gained a lot of attention as an agricultural technique that makes it possible to produce food and related goods in an environmentally friendly way. The general goal of carbon farming is to increase the rate at which CO_2 is removed from the atmosphere and converted into plant biomass and soil organic matter (SOM). This raises the possibility of long-term carbon storage, increases agricultural

yield and enhances soil fertility, all of which lower atmospheric greenhouse gases (GHG) concentrations [16]. Agroforestry, which combines crop and livestock production with forest vegetation, increases net agricultural productivity and food security. It is also well-known for lowering GHG emissions and sequestering carbon, both of which are primarily influenced by the kind of soil, vegetation and land-use techniques [22, 24]. Many environmental factors, including yield, productivity, energy consumption, GHG emission and production processes, can be used to quantify agroforestry ecosystems. This approach of sustainable cultivation practices could be used in sustainable agricultural production, according to researchers [25–27].

An all-encompassing strategy is currently not available for successful carbon farming. Technological solutions and agronomic techniques related to changes in land use are essential components of effective farming [28]. There are five primary carbon farming interventions that are recommended: (1) rewetting and restoring peatlands; (2) establishing and maintaining an agroforestry system; (3) maintaining and improving soil organic carbon (SOC) on mineral soils; (4) managing livestock and manure; and (5) managing nutrients on croplands and grasslands [29, 30]. Peatlands in their natural condition actively collect and store vast amounts of carbon, however, a large portion of Europe's peatland has been drained and degraded, which is releasing carbon into the atmosphere. In terms of greenhouse gas emissions from drained peatlands, the EU is the second-largest emitter in the world (220 Mt CO₂eq/year), accounting for 10% of greenhouse gas emissions from European agriculture and roughly 5% of total EU GHG emissions in 2017. Germany, Finland, Poland, Romania, Ireland, UK, Sweden, Latvia, Lithuania and Netherlands are the nations that emit the most peatlands [31, 32]. A viable carbon farming strategy in locations with a significant amount of cropland on peat soils is rehabilitation and rewetting of drained peatland. The integration of woody vegetation, such as trees or shrubs, with crop and/or livestock farming practices on a same piece of land is known as agroforestry. The permanent crop systems in southeast Europe, the vast stretches of Dehesa and Montado in the drylands of Spain and Portugal, and the wooded pastures and bocage landscapes farther north are a few examples [31]. While these long-standing farming practices still store carbon, many of them run the risk of degrading or losing their woody components, which would release more carbon into the atmosphere. Agroforestry has the ability to help mitigate climate change by preserving and reviving these ancient systems and establishing new agroforestry on arable and grassland farms [29, 33, 34].

SOC has been shown to have positive effects on agricultural productivity, soil quality, and climate adaptation as well as mitigation. Given that the expected yearly emissions from mineral soils under cropland in the EU are 27 Mt CO₂eq, maintaining current SOC levels is imperative. The likelihood for carbon sequestration in mineral soils is likewise substantial, although it varies greatly at the farm and plot levels due to soil heterogeneity, climate, SOC levels, and management techniques. Techniques for managing the land that raise SOC levels include agroforestry, cover crops, better crop rotations and avoiding conversion to farmland or grassland [31, 35, 36]. In a study on European carbon farming initiatives discovered that 81% of agricultural emissions in Europe are attributable to the livestock industry. Cost-effective reductions in livestock greenhouse gas emissions can be achieved on farms by adjustments to crop management, animal waste management, energy usage, fertilizer use and herd management and feeding practices. Computer programs known as “whole farm carbon audit tools” use input data that summarizes the farm's management components to determine the farm's greenhouse gas emissions as well as other indicators like nitrogen balance. Incentives for climate action aimed at bringing GHG reductions below the current baseline level can be obtained through a carbon audit of animal farms [19, 20, 37–39].

Four different types of grassland conversion and management that help to sequester carbon on grasslands through changes in SOC were the focus of a case study conducted by the European Union for the establishment and implementation of result-based carbon farming mechanisms in the EU. The following included maintaining the current grasslands, converting “fallow/set-aside” areas to grasslands; substituting grasslands for annual cropland, including economically marginal arable land, such as sloping land or shallow soils, for grassland management; and avoiding emissions from avoiding converting grasslands to arable land on cultivable soils [24–28, 40–43]. Certain farming practices and geographical areas are better suited for different mitigating strategies. Traditional tillage, more effective fertilizer usage, bio-fertilizers, mulching, cover crops, improved crop rotations, land-use change, peatland restoration, growing agroforestry systems and changes in the kind and location of production are the main strategies employed [29, 33].

Carbon farming frequently offers other environmental or financial advantages, such as the preservation of biodiversity or financial savings for farmers, in addition to reducing the effects of climate change. Numerous co-benefits for the environment and agricultural sustainability can be obtained from it, such as enhanced yield stability and crop resilience against climate change, as well as more effective utilization of crop nutrients and livestock feeding schedules [44, 45]. The development of technological solutions can also lower the intensity of greenhouse gas emissions per unit of output and

increase resource efficiency. Examples of these include the generation of biogas from agricultural waste, modifications to livestock housing, and the use of nitrification inhibitors [46].

4 Conservation farming/agriculture

Crop monocultures and deep tillage with layer inversion have been used to exploit agricultural soils, which has led to a gradual deterioration of the soil's structure, compaction and loss of organic matter. These harmful advancements have increased soil water and wind erosion, CO₂ emissions and adverse cascade consequences on soil fertility and biota [47]. Conservation agriculture (CA) is one of the most advanced alternative management approaches to traditional agriculture that aims to maintain agricultural systems [48]. "A sustainable agricultural production system for the protection of water and agricultural soil that integrates agronomic, environmental, and economic aspects" is how the Food and Agriculture Organization of the United Nations defines CA [49]. The three pillars of CA are: permanent soil organic cover with crop residues and/or cover crops; minimum mechanical soil disturbance through conservation tillage (i.e., no tillage, minimum tillage); and diversified farming through the rotation and associations including at least three distinct crops (which comprises a legume crop) [50].

Adopting any of the three CA principles has positive impacts on soil organic carbon (SOC), as has been shown globally. Conservation Agriculture enhances the physical, chemical, and biological qualities of soil, which are essential for preserving soil condition and health [51]. With its ability to mitigate climate change and its well-known agro-environmental benefits, the rise in SOC content is a great way for assessing how effective a particular CA approach is.

Because CA has the ability to improve crop output while still preserving soil, it continues to gain popularity worldwide. Also it can enhance the state of degraded soil at the landscape level, which may support the region's long-term production. For example, CA methods are being studied for the same reason in several parts of East Africa [52, 53]. This management strategy preserves soil moisture, SOC and SOM while minimizing surface water runoff and soil erosion. Mafongoya et al. (2016) conducted a meta-analysis of several conservation alternatives in Zimbabwe to examine the overall yield of maize in comparison to traditional agricultural methods. For example, with CA techniques such direct seeding, rip-line seeding, and planting basin seeding, yield was determined to be 241 kg ha⁻¹, 258 kg ha⁻¹ and 445 kg ha⁻¹, respectively [54]. According to Sapkota et al. [55], CA was shown to minimize around one-fourth of the production costs for the Indo-Gangetic Plain (IGP) region and enhance irrigation water productivity by at least 65% or more. It also lowered canopy temperature by 2.5 °C to survive climate change and lower GHG emissions into the atmosphere. In the same manner [55], CA is gaining popularity in nations like Brazil, Paraguay and Africa. By choosing CA over conventional farming techniques, the UN-FAO has already documented a rise in the average yield trend for Brazil's 3 main crops, wheat, soybeans, and maize, during an 8-year period [56].

5 Integrated pest management (IPM)

Pathogens, weeds and insects represent a hurdle to achieving sustainable development and global food safety since they drastically decrease agricultural production worldwide. In terms of food security, yield losses from pests might be equal to the quantity of food required to feed about a billion people [57]. In order to reduce pest damage and prevent the costs and negative consequences of synthetic pesticides, other methods must be employed. But overuse of these pesticides raises other problems, and it is now apparent that these should be avoided [33]. Since its establishment in the late 1950s, integrated pest management or IPM, has been the predominant crop protection paradigm. "Integrated Pest Management means careful evaluation of all accessible pest control techniques and thereafter integrating suitable measures that dissuade the growth of pest populations while limiting pesticides and other strategies to levels that are economically feasible and reduce or minimize risks to human health and the environment" states the Food and Agriculture Organization of the United Nations (FAO) [20]. Integrated Pest Management supports natural pest management methods and the development of productive crops with the least amount of disturbance to agro-ecosystems. It's a disciplined approach that combines numerous pest management techniques into one programme. Reducing reliance on pesticides requires incorporating cultural, biological, genetic, physical, legal and mechanical boundaries [58]. Many solutions have been devised and put into practice to manage the pests known as the South American tomato pinworm, *Tuta absoluta* (Meyrick), which is endangering tomato output worldwide, according to a study undertaken by researchers on pest management techniques. The study showcased the effective application of IPM strategies for tomato crops

in several global regions affected by *T. absoluta* infestations. The authors noted that although pesticide resistance is an increasing issue, the most effective management strategies involve biological control through the release or preservation of arthropod natural enemies and sex pheromone-based biotechnical control [59, 60]. IPM effectiveness may be increased through the use of resistant cultivar breeding, soil fertilization, irrigation and other developing research related to agronomic control. IPM can increase crop yields by lowering pest-related crop losses and minimizing the negative effects of pesticide use on the environment. According to a study, the implementation of IPM techniques in rice cultivation in Asia resulted in a 64% decrease in pesticide use and a 14% increase in output [61]. A study by Saliu et al. [62] revealed that crop rotation, cover crops, conservation tillage, organic farming and IPM are examples of sustainable agricultural techniques that can boost crop yields and improve soil health. By enhancing soil biodiversity, nutrient cycling, and organic matter, these techniques can increase the resilience of agricultural ecosystems [62]. While there could be trade-offs between long-term environmental and social benefits and gains in production in the short term, incorporating different sustainable agriculture practices can help strike a balance between these goals. By utilizing the synergies between these approaches, we can meet the difficulties of feeding an increasing population while protecting natural resources and reducing our influence on the environment. This can lead to a resilient and sustainable global food system [63, 64].

6 Precision farming/agriculture (PA)

Using data and modern technologies, Precision Agriculture (PA) is a management method that addresses temporal and geographical variability in agricultural fields [34]. The amount of food produced globally needs to increase by at least 70% by 2050. This is a challenging project because it progressively depletes the environment and already limited resources [65]. As a result, PA is necessary to ensure sustainability, minimize negative effects on the environment and maximize output while utilizing fewer inputs of all kinds in more efficient ways [66]. The development of GPSs (Global Positioning Systems), GISs (Geographic Information Systems), yield monitors and other data generators at all 3 critical stages of agricultural operations (diagnosis, decision making & performing) during the 1990s gave rise to precision farming. In precision farming, mechanical machinery was limited to carrying out agricultural tasks; human authorization was required for problem-solving and decision-making processes [67].

Precision agriculture involves the collection of data on specific fields and crops by observation, measurement, and sensing using various types of sensors, yield and soil monitors, and remote sensing instruments, such as imagery from crew members, aircraft, satellites, or drones. Therefore, “sensing”—which is the observation of precise data and provision of information on climate conditions, soil conditions, fertilizer requirements, water availability, pest, and disease stressors, among other field parameters—is a crucial management tool of PA [68]. Data on crops and soil conditions are gathered via remote-sensing technology, including drones, airplanes, satellites, and other ground-based sensors. The identification of geographical patterns of plant signatures that coincide with soil properties and pest or disease stressors is made possible with the use of remote sensing. One type of remote sensing data that can show ground truthing is imagery [69]. A global navigation satellite system (GNSS) can offer geographical coordinates and other data that are used to build maps, particularly yield maps and soil maps for management decisions relevant to a certain location. In order to make quantitative and qualitative field production characteristics, yield maps are utilized, which is essential for management decision-making [70]. Variabilities represented on maps can be analyzed to identify productivity-influencing elements and make site-specific field management plans easier to apply. A network of linked objects and technologies is referred to as the Internet of Things (IoT). One of the most significant technology developments in smart farming and precision agriculture is the IoT [68, 71]. Precision agriculture uses data collected by IoT architecture, including agricultural sensors with ICT and UAV (Unmanned Aerial Vehicle). Additionally, the fourth industrial revolution is centered on the expanding IoT and mobile data. With the development of sensors with independent intellectual property rights and smart devices like intelligent tractors, unmanned aerial vehicles and robots that can replace large amounts of manual labor input and perform high-quality tasks while adapting to difficult working conditions, the IoT can be widely used in all areas of precision agriculture [69, 72].

Drones, satellite imaging and ground-based sensors are all used in crop monitoring applications to evaluate plant health, identify diseases and improve watering techniques. With the help of weather forecasts, historical patterns and data-driven projections, AI-driven decision support systems enable farmers to make well-informed decisions that minimize environmental impact and promote resource efficiency. Sustainable farming focusses significantly on resource management, and artificial intelligence is essential to maximizing the use of pesticides, fertilizers and water. AI-powered smart irrigation systems provide accurate and effective water distribution, minimizing water waste

and encouraging water conservation. Farmers may better utilize nutrients and reduce environmental discharge by customizing their fertilization operations with the aid of AI-driven soil condition analysis [73, 74].

7 Impact on soil health

In addition to providing several ecosystem services like maintaining crop production and water quality, a healthy soil also regulates the decomposition and recycling of soil nutrients and absorbs greenhouse gases from the environment. Because the primary determinants of soil health are the diversity and activity of soil microorganisms, soil health and sustainable agriculture are closely interconnected [74]. In order to maintain or improve water and air quality, support plant and animal health and function as a viable living system within ecological and land use constraints, a soil must possess certain qualities, which Doran and Zeiss characterized as soil health [75]. A soil's inherent quality is its state of health. It is acknowledged as a set of traits that categorize and characterize its state of health. On the other hand, soil quality is an external feature of soils that varies depending on how humans intend to use them. It might have to do with how well agriculture produces products for recreation, preserves watersheds, and supports wildlife [76].

Since leaving their aquatic environment to establish themselves on the land, plants and microorganisms have co-evolved for more than 400 million years, resulting in the formation of extremely complex soil-microbe-plant systems that carry out numerous vital biological and ecological functions, such as nutrient cycling, carbon sequestration, soil fertility maintenance, and ecosystem resilience [77]. Because highly chosen crop species are used in various cropping systems for both nutritional and fiber advantages, which also greatly boost the "host effects" on soil microorganisms, these soil-microbe-plant interactions in agriculture are even stronger. Throughout their co-evolution, plants—which are sessile organisms—developed a variety of chemical signaling pathways to influence and regulate the root microbiome. Angon et al. (2023) documented the positive effects of integrated pest management strategies on soil biodiversity [20].

- Improvement of soil structure and fertility

The potential of precision farming technologies for environmentally friendly agriculture was examined by Gawande et al. [21]. They came to the conclusion that precision farming allows farmers to maximize the use of resources like water, fertilizer and pesticides, ensuring that crops receive the exact amount needed, which increases productivity and improves soil texture, water-holding capacity and biodiversity [21]. The physical characteristics of soil include its structure, texture, bulk density, porosity and ability to hold water. Research has shown that organic farming and other sustainable farming methods have a positive impact on the physical characteristics of soil, including its structure, water-holding capacity, aeration and temperature. According to Papadopoulos et al., organic management can enhance the porosity, organic matter content, and soil structure [78]. One key element of sustainable farming that both directly and indirectly affects the physical structure of the soil is crop rotation, a carbon farming strategy. The change of soil structure is directly influenced by the organic matter that accumulates in the soil during the lean phase [79]. Modifying the soil structure is also aided by the architectural form of the various root systems of the various crops that are part of the crop rotation. By covering the soil's surface with organic materials, mulching the soil produces a soft, pulverized and humid soil, which in turn fosters the growth of beneficial bacteria that help the soil's bulk density and porosity. Increasing the amount of organic matter in the soil through sustainable farming practices is a fundamental prerequisite for enhancing soil health. This organic stuff improves the soil's ability to hold onto moisture [80]. In comparison to non-organic plots, sustainable agricultural techniques (bio-fertilizers, crop rotation, conventional tillage, varied cropping, etc.) managed plots showed significant differences and higher levels of soil organic carbon, carbon stocks, and carbon sequestration rate [23, 81]. Without a doubt, it has a significant role in regulating the C: N ratio, soil moisture, microbial activity, total and accessible N, N mineralization, and soil texture. Surprisingly, a number of studies have found that soil amended with organic matter has more accessible nitrogen than soil fertilized with inorganic fertilizers [82]. This is mostly because the slower and more consistent mineralization rates of organic soil prevent nitrogen leaching. The decomposing matter's humus percentage and organic acids are more effective at liberating phosphorus from the soil and lowering its fixation. Micronutrient availability for the plant is additionally guaranteed by nutrient delivery from organic sources [83].

- Enhancement of soil biodiversity

The term “biodiversity in soil” describes the range of taxonomic taxa that can be found in soil, such as nematodes, earthworms, bacteria, fungi, and protozoa. Modern agriculture’s intensification and spread have put biodiversity at risk globally and numerous studies have revealed that during the last 40 years, it has decreased the diversity and abundance of a wide range of plant and invertebrate taxa [77, 84]. Sustainable agricultural methods, on the other hand, support biodiversity preservation. Following a 21-year investigation into the agro-economic and agro-ecological performance of conventional, biodynamic, and bioorganic farming systems in Central Europe, it was found that organic farming produced greater soil fertility and biodiversity than conventional farming [73, 85].

To maintain soil production, microbial biomass and microbial activity in the soil are essential. A balanced ratio of microbial biomass and activity in the soil is necessary to guarantee the consistent delivery of nutrients to plants. According to reports, conventional farming has increased microbial activity and biomass by 30–100% and 20–30%, respectively [73, 86]. Greater microbial activity and soil N supply are guaranteed by soil with a high organic matter level compared to soil with a low organic matter concentration (Table 1) [87]. Furthermore, soil organic matter has the ability to absorb CO₂ from the atmosphere, raising the carbon content of the soil and promoting microbial respiration and biomass [74]. Additionally, it is widely known that soil that has been organically managed and supplemented with a variety of advantageous microorganisms, such as arbuscular mycorrhizal fungus, improves crop nutrition and lowers the incidence of soil-borne illnesses [86].

8 Alignment with sustainable development goals (SDGs)

The SDGs of the United Nations, particularly SDG #2 (zero hunger), #3 (good health and wellbeing), #6 (clean water & sanitation), #13 (climate action), and #15 (life on land), are interconnected with the issues of climate change, soil degradation and food security [88]. These goals, along with the Millennium Development Goals and Agenda 21, emphasize achieving food security, ending hunger and eradicating poverty [89]. Under SDG #2, improving food quantity and quality through soil health management plays a crucial role in eradicating hunger. Sustainable agriculture, which includes maintaining soil organic carbon (SOC), is key to achieving these objectives as demonstrated in Fig. 3. Enhancing SOC through soil carbon farming supports sustainable agriculture by increasing food yield without compromising environmental goods and services, thus improving soil quality and agricultural sustainability [90].

Improving soil health is vital for achieving zero hunger and sustaining terrestrial life, as it impacts human, animal and ecosystem health. Enhancing soil organic matter (SOM) is central to this effort, as it increases soil carbon storage, improves aeration, water-holding capacity & nitrogen exchange and maintains buffering capacity [91]. Agroforestry, which integrates trees with agriculture, can enhance nutrition and well-being, particularly in rural areas. For instance, fruit tree production on farms can significantly impact child nutrition. Proven soil management techniques that increase SOC content can potentially quadruple staple food crop production in Sub-Saharan Africa, enhance nutritional quality and reduce emissions from farming. These techniques also improve soil characteristics and water retention, resulting in healthier, more productive soils, especially in marginal conditions [92, 93].

Conservation agriculture, which includes minimal tillage, permanent soil cover and crop rotations, addresses low agricultural productivity and soil degradation. CA benefits include improved SOM, better water harvesting, increased yields and enhanced food security [94]. Studies in Zimbabwe and Zambia have shown that CA with residue retention improves soil moisture and water infiltration [76, 95]. Agroforestry, particularly with leguminous trees, can decrease soil carbon losses due to erosion and increase soil carbon storage, moisture retention and drought resistance [96]. Climate-smart agriculture (CSA) encompasses practices like low tillage, mulching, intercropping and integrated crop-livestock management, aiming to create resilient agricultural systems that adapt to climate variability, mitigate GHG emissions and ensure sustainable productivity [97]. For example, using green manures in Cuba improved soil properties and enhanced crop yields [98]. Integrated soil management techniques that increase SOM, control residual nutrients and promote sustainable agriculture are essential for achieving the SDGs, including ending hunger and poverty and improving health and climate resilience [99–101].

Table 1 Positive Outcomes of Sustainable Farming Practices and SDG Goals (SDG number 13: Climate action; 15: Life on land; 2: Zero hunger; 3: Good health and well-being; 12: Responsible consumption and production)

Serial no	Farming practice	Impact on agriculture	SDG number	Reference
1	Carbon sequestering farming	Increases soil organic carbon, enhances soil fertility, reduces greenhouse gas emissions	13, 15	[13, 58, 59]
2	Conservation farming	Improves soil structure, increases water retention, enhances biodiversity	2, 13, 15	[14, 26]
3	IPM farming	Reduces pesticide use, enhances biodiversity, maintains crop yields	2, 3, 15	[16, 59]
4	Precision farming	Optimizes input use, increases crop yields, improves resource-use efficiency	2, 12, 13	[40, 45, 60]

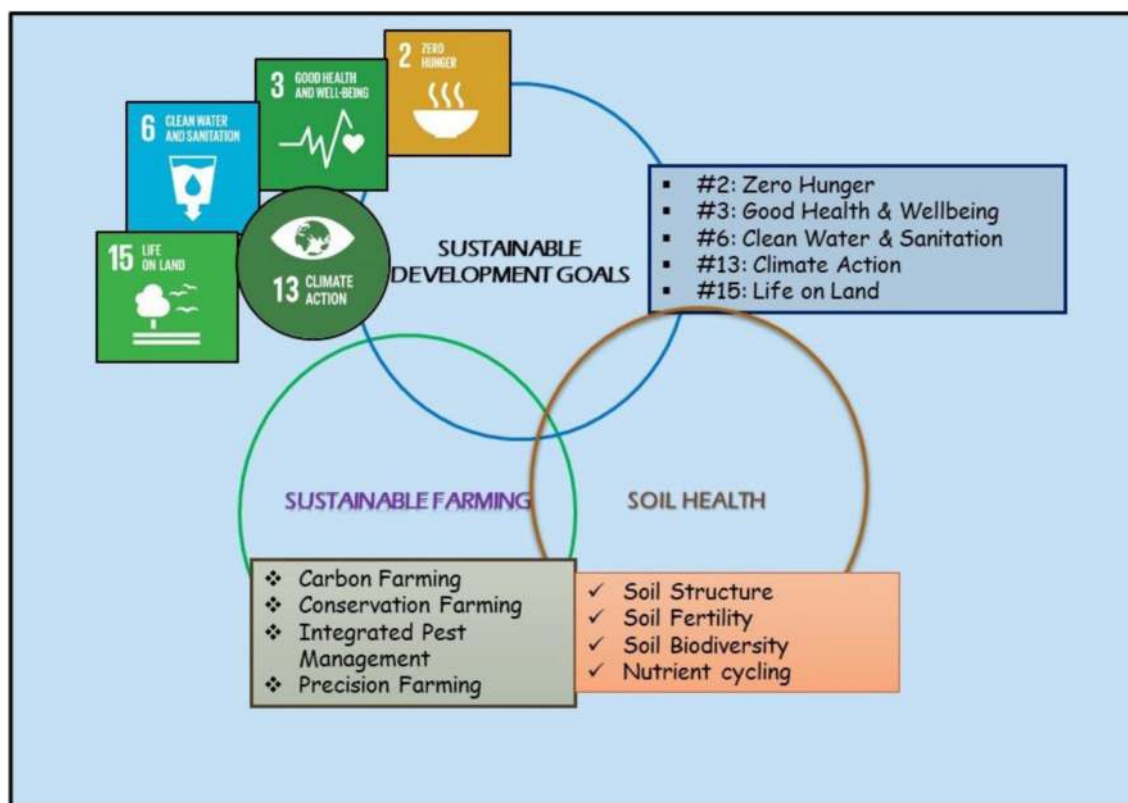


Fig. 3 The interconnection of sustainable farming with key environmental factors: alignment with Sustainable Development Goals (SDGs) and enhancement of soil health for sustainable agriculture

9 D. Conclusion and future directions

Sustainable farming practices are essential to maintaining and improving soil health, which is crucial for the long-term sustainability of agriculture. These practices, including integrated pest management, precision farming, carbon farming and conservation agriculture, offer innovative solutions to the challenges posed by conventional agricultural methods. They aim to enhance soil fertility, structure and biodiversity, thereby supporting higher agricultural productivity and resilience. The implementation of these practices not only addresses immediate food security concerns but also help mitigate the adverse effects of climate change by sequestering carbon and reducing GHG emissions and improving water quality. As we get closer to the 2030 deadline for attaining the SDGs, the importance of implementing sustainable farming techniques cannot be emphasized. There are only 6 years left, therefore we need to take action urgently to change agricultural systems into sustainable role models that can both meet the world's food needs and maintain environmental integrity. The importance of sustainable agriculture in achieving important SDGs including ending world hunger, combating climate change and preserving land is highlighted by this review. The possibility of achieving these worldwide goals is compromised if these techniques are not put into practice in the remaining period. It's time to put things into action; a swift transition to sustainable agriculture is necessary to ensure a secure healthy future for everybody.

Future research should focus on developing and optimizing these sustainable practices across different agro-ecological zones, integrating modern technologies and promoting policies that support sustainable farming at local and global levels. It should emphasize the development of advanced sustainable agricultural techniques and innovative soil health management strategies to address the dual challenges of global food security and climate change.

- Research on advanced precision agriculture technologies, including IoT and AI, can optimize resource use, increase crop yield, and improve soil health.

- Long-term studies on the ecological impacts of sustainable farming practices can provide insights into their effectiveness and guide policy development.
- Expanding sustainable agriculture education and outreach programs can promote the adoption of these practices, enhancing global food security and environmental sustainability.

Author contributions P.S. conceived the idea for the review paper and led the overall direction and planning. P.S. conducted the literature review, collected data and contributed to the writing of the manuscript. N.T. provided critical feedback, edited the manuscript and contributed to the interpretation and presentation of the findings. All authors have read and approved the final version of the manuscript. All authors reviewed the manuscript. "P.S. contributed to the idea framing and review writing whereas the final drafting was done by N.T.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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